

Annotated
Version

MACHINE LEARNING COURSE · CS-433

Regression

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EPFL

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The regression problem

What is regression?

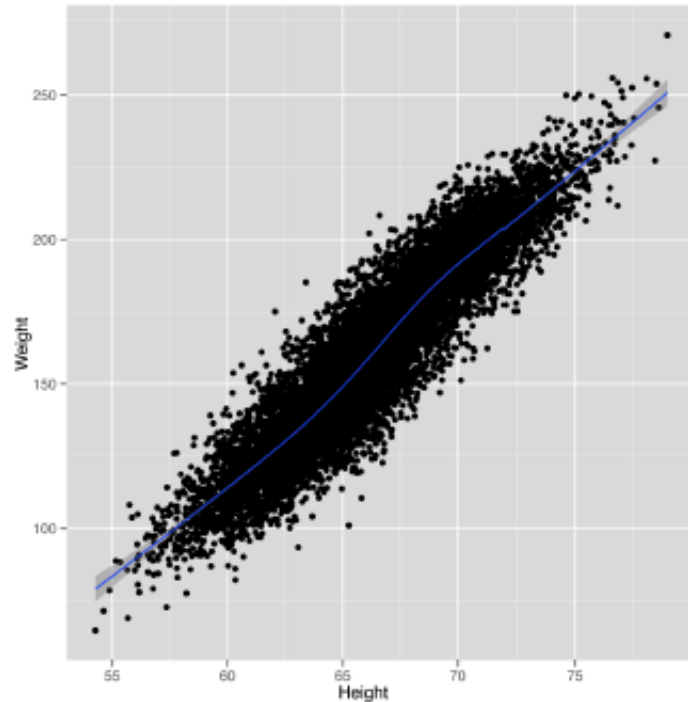
Regression is to relate input variables to the output variable, to either predict outputs for new inputs and/or to understand the effect of the input on the output.

Dataset for regression

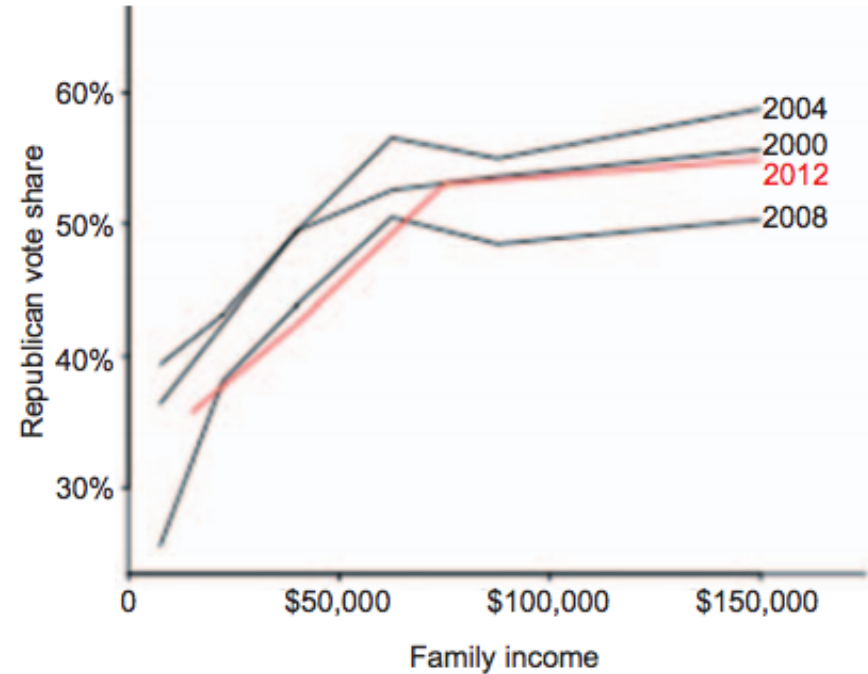
In regression, **data** consists of pairs (x_n, y_n) , where y_n is the n 'th **output** and x_n is a vector of D **inputs**.

The number of pairs N is the **data-size** and D is the **dimensionality**.

Examples of regression

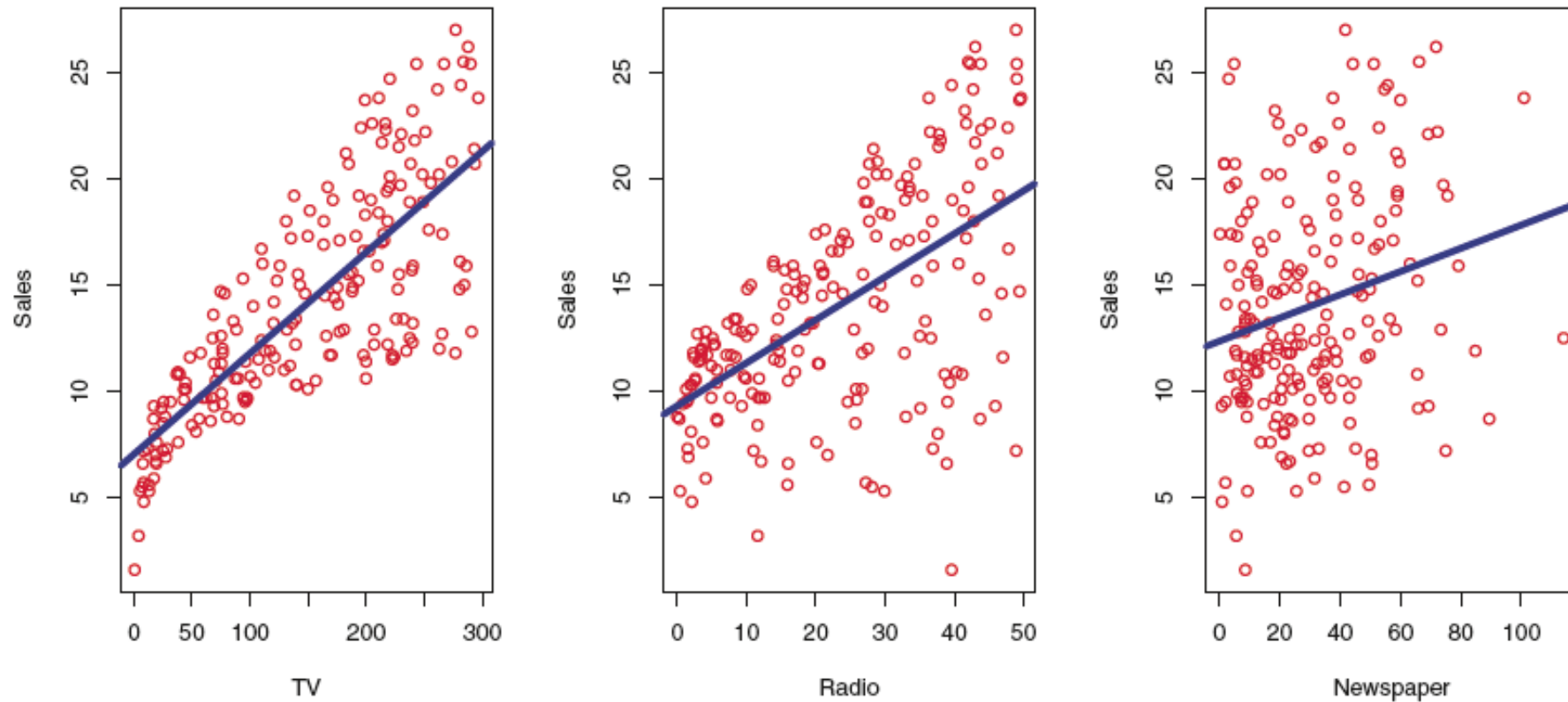


(a) Height is correlated with weight. Taken from “Machine Learning for Hackers”.



(b) Do rich people vote for Republicans? Taken from Avi Feller et al., 2013, “Red state / blue state in 2012 elections”.

Examples of regression



(c) How does advertisement in TV, radio and newspaper affect sales? Taken from the book “An Introduction to Statistical Learning”.

Two goals of regression

In **prediction**, we wish to predict the output for a new input vector, e.g. what is the weight of a person who is 170 cm tall?

In **interpretation**, we wish to understand the effect of inputs on output, e.g. are taller people heavier too?

The regression function

For both the goals, we need to find a function that approximates the output “well enough” given inputs.

$$y_n \approx f(\mathbf{x}_n), \quad \text{for all } n$$

Everything that follows is about choosing this f , and then fitting it.

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Linear regression

We start with the simplest form for f that is still useful: a linear function.

What is it, and why start here?

Linear regression is a **model** that assumes a linear relationship between inputs and the output — exactly what the height-vs-weight example already suggested.

Why learn about linear regression?

Plenty of reasons: simple, easy to understand, most widely used, easily generalized to non-linear models.

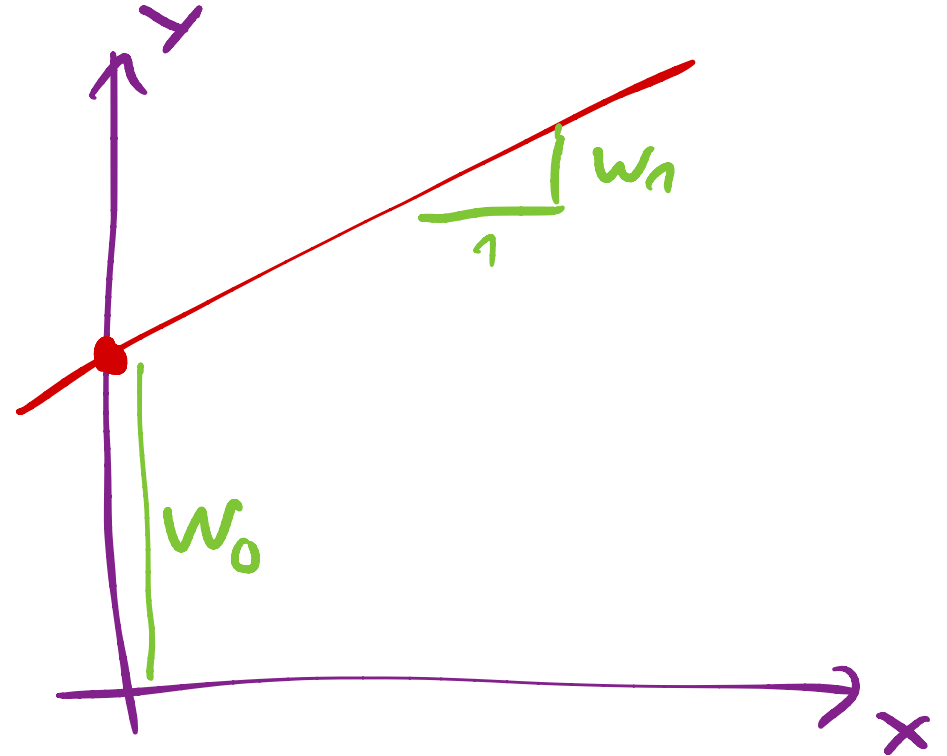
Most importantly, you can learn almost all fundamental concepts of ML with regression alone.

Univariate linear regression

With only one input dimension, we get univariate linear regression.

$$y_n \approx f(\mathbf{x}_n) \stackrel{//}{=} f_w(w) := w_0 + w_1 x_{n1}$$

Here, $w = (w_0, w_1)$ are the two parameters of the model. They describe f .

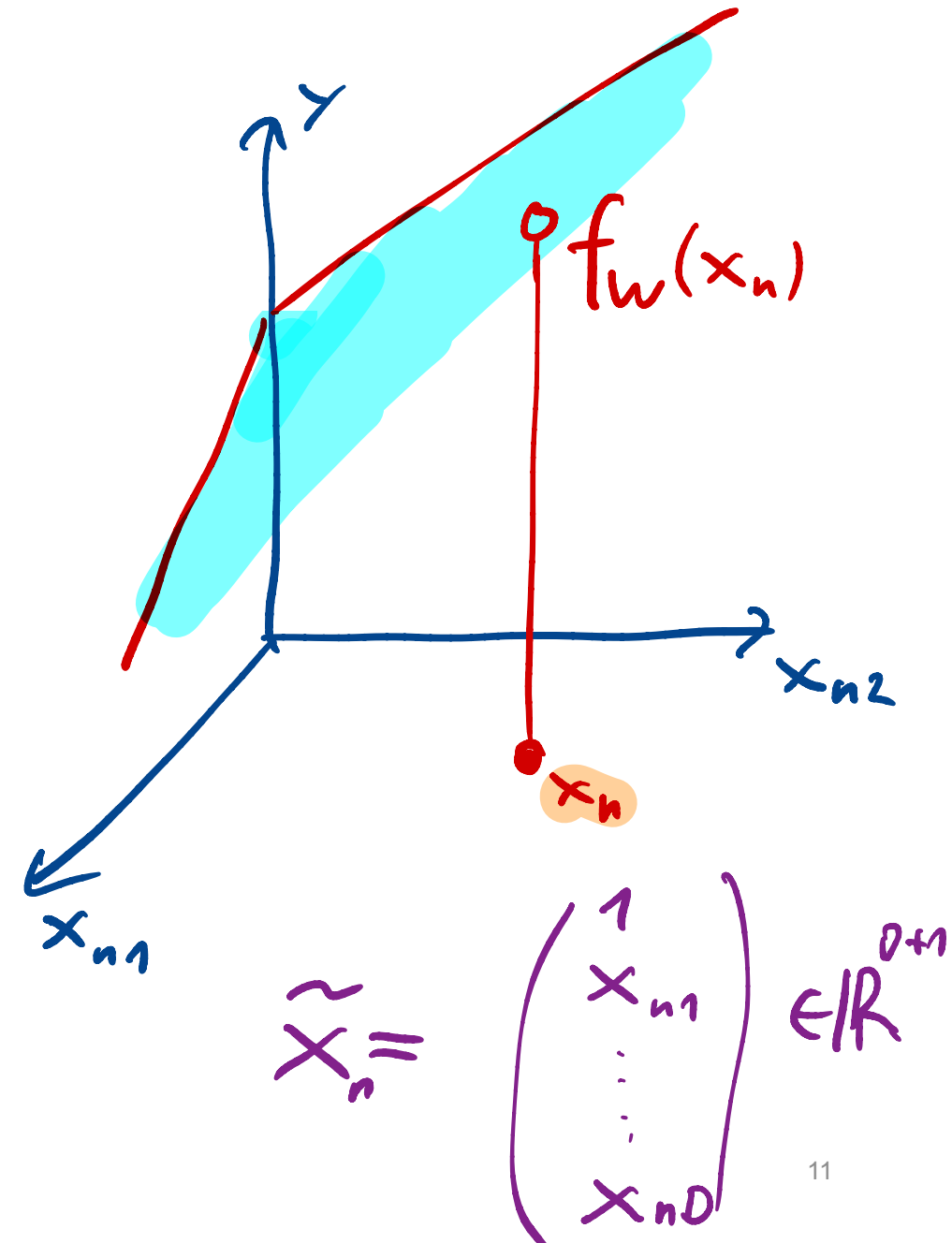


Multivariate linear regression

If our data has multiple input dimensions, we obtain multivariate linear regression.

$$\begin{aligned} y_n &\approx f(\mathbf{x}_n) = f_w(\mathbf{x}_n) \\ &:= w_0 + w_1 x_{n1} + \dots + w_D x_{nD} \\ &= w_0 + \mathbf{x}_n^\top \begin{pmatrix} w_1 \\ \vdots \\ w_D \end{pmatrix} \\ &=: \tilde{\mathbf{x}}_n^\top \tilde{\mathbf{w}} \end{aligned}$$

Note that we add a tilde over the input vector, and also the weights, to indicate they now contain the additional offset term (a.k.a. bias term).



Learning / Estimation / Fitting

Given data, we would like to find $\tilde{w} = [w_0, w_1, \dots, w_D]$. This is called **learning** or **estimating** the parameters, or **fitting** the model.

To do so, we need an optimization algorithm, which we will discuss in the chapter after the next.

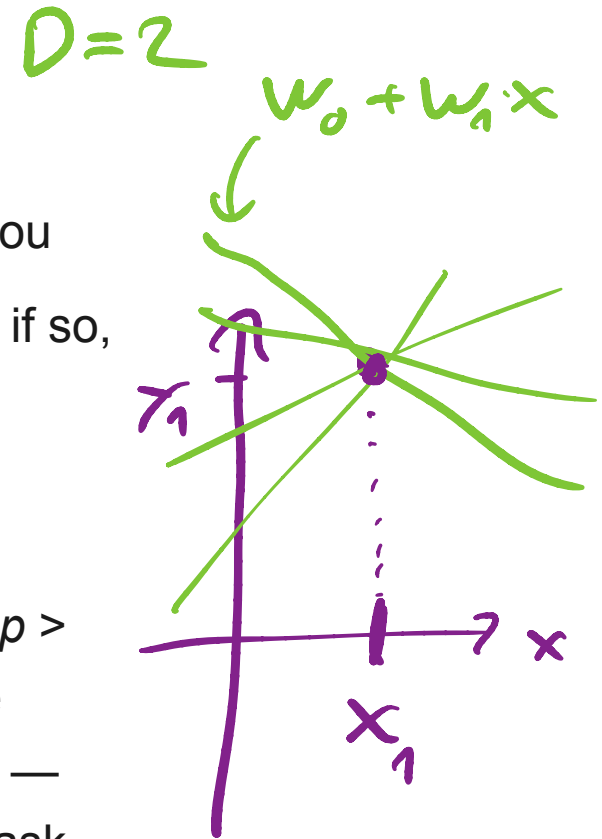
Overparameterization — the $D > N$ issue

Consider a simple situation: you have $N = 1$ and you want to fit $y_1 \approx w_0 + w_1 x_{11}$, i.e. you want to find $w = (w_0, w_1)$ given one pair (y_1, x_{11}) . Is it possible to find such a line, and if so, is it uniquely determined?

underdetermined

This identification issue is related to the $D > N$ problem (in statistics typically named $p > n$): the number of parameters exceeds the number of data examples. We have more variables than data information, which for many models — such as linear regression — makes the task *under-determined*. We say the model is *over-parameterized* for the task.

Regularization is a way to avoid this identification issue, which we will learn later. If we ignore the issue, overparameterization can also have advantages, e.g. on the training dynamics.



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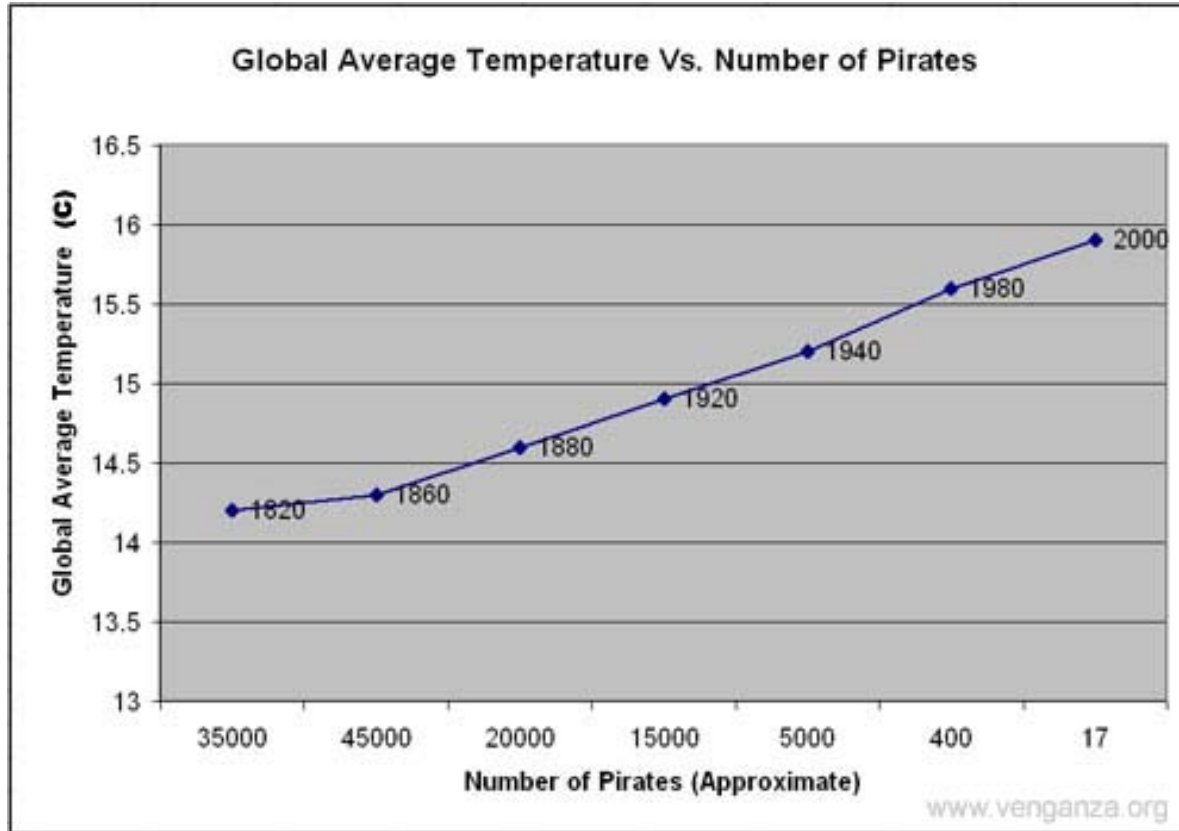
Additional notes

(Not exam material)



Correlation $\neq$ causation

Regression finds correlation, not a causal relationship, so interpret your results with caution.



This image is taken from www.venganza.org. You can see many more examples at Tyler Vigen's [Spurious correlations page](#).

Machine learning jargon for regression

Input variables are also known as features, covariates, independent variables, explanatory variables, exogenous variables, predictors, regressors.

Output variables are also known as target, label, response, outcome, dependent variable, endogenous variables, measured variable, regressands.

Prediction vs. interpretation

Some questions to think about: are these prediction tasks or interpretation tasks?

1. What is the life expectancy of a person who has been smoking for 10 years?
2. Does smoking cause cancer?
3. When the number of packs a smoker smokes per day doubles, their life span gets cut in half?
4. A massive-scale earthquake will occur in California within the next 30 years.
5. More than 300 bird species in North America could reduce their habitat by half or more by 2080.

Alternative without an offset term

Above we used $D + 1$ model parameters to fit data of dimension D . An alternative, often used in practice for high-dimensional data, is to ignore the offset term w_0 .

$$\begin{aligned} y_n \approx f(\mathbf{x}_n) &:= w_1 x_{n1} + \dots + w_D x_{nD} \\ &= \mathbf{x}_n^\top \mathbf{w} \end{aligned}$$

In this case we have just D parameters to learn, instead of $D + 1$.

As a warning: for some ML models the number of weight parameters (elements of w) can be very different from D (the dimension of the input data, i.e. the number of features). For a model where they do not coincide, think of a neural network — more details later.

Matrix multiplication

To go any further, one must revise matrix multiplication. Remember that multiplying an $M \times N$ matrix with an $N \times D$ matrix results in an $M \times D$ matrix.

Also, two matrices of size $M \times N_1$ and $N_2 \times M$ can only be multiplied when $N_1 = N_2$.